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ACADEMY

IGCSE Maths Paper 2 Full Revision Paper

Cambridge IGCSE Mathematics 0580, Paper 2 Non-calculator (Extended). A complete 100-mark paper built on the weighting of the 2025 specimen and the 2025 live papers, one question per page, worked live on screen across six videos. Blank workspace sized for annotation in Xournal++.

26	100	2:00	6	16 : 9
questions	marks	exam time	videos · 1h 45m	254 × 143 mm

PREPARED FOR	DATE	SYLLABUS	NEXT SITTING
Nur Nabi	**28 September 2026**	**0580 (2025–2027) · Extended**	**Paper 2 · Thu 8 October 2026**

READ THIS BEFORE YOU RECORD

How to use this paper

Two ways to use it

- **Students:** sit it as a mock first — 2 hours, no calculator, all questions — then watch the videos to mark it.
- **Videos:** one video per section. Each page carries its time budget top-right: one minute per mark, the same pace the real exam allows.

In Xournal++

- Open this PDF as the **background** (Annotate PDF), so text stays crisp at any zoom.
- Page size is **254 × 143 mm**, exactly 16:9 — full-page view fills a 1920 × 1080 frame.
- Dark blue or black pen for working, **red only for the final answer** and the mark it earns.
- Say the method mark out loud as you earn it: "that's M1 for the common denominator".

Everything here is original. The questions are new, written against the 0580 syllabus and modelled on the style and weighting of real Paper 2 questions. No Cambridge question is reproduced, so the videos and the worked-solutions PDF are safe to publish and monetise.

Say this in every video. This paper is a practice paper, not a prediction. No one knows what will be on the real paper — the topic mix follows the syllabus and recent papers, so every topic here is worth knowing.

Paper 2 rules (syllabus p. 59): 2 hours · 100 marks · no calculator · answer all questions · Extended content except E1.14 (Using a calculator). The List of formulas is on page 2 of the real paper.

HOW THIS PAPER WAS BUILT

Marks by topic — the blueprint

Weighting taken from four real Paper 2s: the **2025 specimen**, **0580/21** and **0580/22 June 2025**, and **0580/22 November 2025**. "4/4" means the topic was examined in all four papers. Algebra carries about a third of the marks, as it does in the real papers.

TOPIC AREA	MARKS	WHAT GETS TESTED	SEEN IN RECENT PAPERS
Number	19	Surds; fractional and negative indices; fraction working; standard form; recurring decimals; prime factors; percentages	surds 4/4 · indices 4/4 · fractions 3/4 · standard form 2/4 · recurring 2/4 · prime factors 2/4 · percentages 2/4
Algebra and graphs	33	Quadratics and completing the square; functions; differentiation; algebraic fractions; expanding; proportion; sequences; rearranging; simultaneous	quadratics 4/4 · functions 3/4 · differentiation 3/4 · algebraic fractions 3/4 · expanding 2/4 · proportion 2/4 · sequences 2/4
Coordinate geometry	6	Midpoint, length, gradient, equation of a line, perpendicular lines	2/4
Geometry	8	Angles and polygons; circle theorems with reasons; similarity	angles/polygons 4/4 · circle theorems 3/4 · similarity 2/4
Mensuration	7	Arc, sector, cone and sphere in terms of π	4/4
Trigonometry	6	Exact values; trig equations for 0°–360°	4/4
Transformations and vectors	7	Describing transformations; column vectors; magnitude; vector geometry	vectors 3/4 · transformations 2/4
Probability	4	Tree diagrams, without replacement, Venn diagrams	3/4
Statistics	10	Cumulative frequency; histograms; estimated mean; averages	4/4 (cumulative frequency 3/4 · histogram 2/4)
Total	**100**	26 questions, rising in difficulty inside each section	

SYLLABUS CHECK

What this paper covers

Q	CODE	TOPIC	MK
Q01	E1.1	Prime factors and square numbers	3
Q02	E1.4 / E1.6	Recurring decimals and fractions	4
Q03	E1.7 / E2.4	Negative and fractional indices	4
Q04	E1.8	Standard form	2
Q05	E1.18	Surds	4
Q06	E1.13	Reverse percentages	2
Q07	E2.2	Expanding three brackets	3
Q08	E2.3	Algebraic fractions	3
Q09	E2.5 / E2.11	Completing the square	4
Q10	E2.5	Changing the subject	3
Q11	E2.5	Simultaneous equations — one linear, one non-linear	4
Q12	E2.7	Linear sequences	2
Q13	E2.8	Direct proportion	3

Q	CODE	TOPIC	MK
Q14	E2.13	Functions — composite and inverse	6
Q15	E2.12	Differentiation and turning points	5
Q16	E3.4 – E3.7	Midpoint, length and perpendicular lines	6
Q17	E4.6	Angles in regular polygons	2
Q18	E4.7	Circle theorems	3
Q19	E4.4	Similar solids — area and volume	3
Q20	E5.3 / E5.4	Sectors and cones	7
Q21	E6.2 – E6.4	Exact trigonometric values and trig equations	6
Q22	E7.1	Transformations — negative scale factor	3
Q23	E7.2 – E7.4	Vector geometry	4
Q24	E8.3	Combined events — without replacement	4
Q25	E9.3 / E9.7	Grouped data — estimated mean and histogram	6
Q26	E9.6	Cumulative frequency	4

SYLLABUS CHECK

Not in this paper — and where to find it

> A 100-mark paper cannot test all 72 Extended subtopics, and neither can the real one. These gaps are covered by your **All of IGCSE Maths in 10 Minutes** deck. Point viewers to it in the description.

CODE	TOPIC	WHERE IT IS COVERED
E1.2	Sets and Venn diagrams	10-minute deck Q02
E1.9 / E1.10	Estimation and bounds	10-minute deck Q06
E1.11 / E1.12	Ratio, proportion and rates	10-minute deck Q07
E1.17	Exponential growth and decay	10-minute deck Q09 (mostly Paper 4)
E2.6	Inequalities and regions	10-minute deck Q15
E2.7	Quadratic sequences	10-minute deck Q16
E2.9	Speed–time and distance–time graphs	10-minute deck Q18
E2.10 / E2.11	Graphs of functions, asymptotes, sketching	10-minute deck Q19
E6.5 / E6.6	Sine rule, cosine rule, 3D trig	10-minute deck Q31–32 (mostly Paper 4)
E8.2 / E8.4	Relative frequency, conditional probability	10-minute deck Q35–36
E4.2 / E4.3	Constructions, scale drawing, bearings	Needs instruments — practise on paper
E1.14	Using a calculator	Excluded from Paper 2 by the syllabus

RECORDING PLAN

Six videos, 15–21 minutes each

VIDEO	SECTION	QUESTIONS	MARKS	RUNTIME
1	**Number**	Q1–Q6	19	19:50
2	**Algebra I — manipulation and equations**	Q7–Q11	17	17:50
3	**Algebra II — sequences, proportion, functions, calculus**	Q12–Q15	16	16:50
4	**Coordinate geometry and geometry**	Q16–Q19	14	14:50
5	**Mensuration, trigonometry, transformations and vectors**	Q20–Q23	20	20:50
6	**Probability and statistics**	Q24–Q26	14	14:50
	Whole paper	Q1–Q26	**100**	1:45:00

Runtime = 0:30 intro + 1 minute per mark + 0:20 outro. Publish all six as one playlist, “IGCSE Maths Paper 2 — Full Revision Paper”, and put the blank paper PDF in every description.

CHAPTER MARKERS

YouTube chapters — paste into each description

Video 1 · Number

START	CHAPTER	BUDGET
0:00	Intro — what this video covers	
0:30	Q1 Prime factors and square numbers	3:00
3:30	Q2 Recurring decimals and fractions	4:00
7:30	Q3 Negative and fractional indices	4:00
11:30	Q4 Standard form	2:00
13:30	Q5 Surds	4:00
17:30	Q6 Reverse percentages	2:00
19:30	Outro — next video	0:20

Video 2 · Algebra I

START	CHAPTER	BUDGET
0:00	Intro — what this video covers	
0:30	Q7 Expanding three brackets	3:00
3:30	Q8 Algebraic fractions	3:00
6:30	Q9 Completing the square	4:00
10:30	Q10 Changing the subject	3:00
13:30	Q11 Simultaneous equations — one linear, one non-linear	4:00
17:30	Outro — next video	0:20

CHAPTER MARKERS

YouTube chapters — paste into each description

Video 3 · Algebra II

START	CHAPTER	BUDGET
0:00	Intro — what this video covers	
0:30	Q12 Linear sequences	2:00
2:30	Q13 Direct proportion	3:00
5:30	Q14 Functions — composite and inverse	6:00
11:30	Q15 Differentiation and turning points	5:00
16:30	Outro — next video	0:20

Video 4 · Coordinate geometry & geometry

START	CHAPTER	BUDGET
0:00	Intro — what this video covers	
0:30	Q16 Midpoint, length and perpendicular lines	6:00
6:30	Q17 Angles in regular polygons	2:00
8:30	Q18 Circle theorems	3:00
11:30	Q19 Similar solids — area and volume	3:00
14:30	Outro — next video	0:20

CHAPTER MARKERS

YouTube chapters — paste into each description

Video 5 · Mensuration, trig & vectors

START	CHAPTER	BUDGET
0:00	Intro — what this video covers	
0:30	Q20 Sectors and cones	7:00
7:30	Q21 Exact trigonometric values and trig equations	6:00
13:30	Q22 Transformations — negative scale factor	3:00
16:30	Q23 Vector geometry	4:00
20:30	Outro — next video	0:20

Video 6 · Probability and statistics

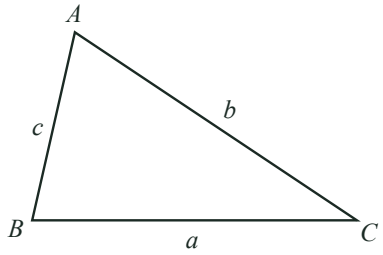
START	CHAPTER	BUDGET
0:00	Intro — what this video covers	
0:30	Q24 Combined events — without replacement	4:00
4:30	Q25 Grouped data — estimated mean and histogram	6:00
10:30	Q26 Cumulative frequency	4:00
14:30	Outro — next video	0:20

List of formulas

Area, A, of triangle, base b, height h.	$A = \frac{1}{2}bh$
Area, A, of circle of radius r.	$A = \pi r^2$
Circumference, C, of circle of radius r.	$C = 2\pi r$
Curved surface area, A, of cylinder of radius r, height h.	$A = 2\pi rh$
Curved surface area, A, of cone of radius r, sloping edge l.	$A = \pi rl$
Surface area, A, of sphere of radius r.	$A = 4\pi r^2$
Volume, V, of prism, cross-sectional area A, length l.	$V = Al$
Volume, V, of pyramid, base area A, height h.	$V = \frac{1}{3}Ah$
Volume, V, of cylinder of radius r, height h.	$V = \pi r^2 h$
Volume, V, of cone of radius r, height h.	$V = \frac{1}{3}\pi r^2 h$
Volume, V, of sphere of radius r.	$V = \frac{4}{3}\pi r^3$

For the equation $ax^2 + bx + c = 0$, where $a \neq 0$ $\quad x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

For the triangle shown,



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$\text{Area} = \frac{1}{2}ab\sin C$$

> Given in the exam — you do **not** need to memorise these. Everything else (e.g. arc length, sector area, trapezium, Pythagoras, compound interest) must be known.

1

Number

Video 1 of 6 · Questions 1–6 · 19 marks · E1.1 – E1.18 · 19:50 target runtime

Question 1

Prime factors and square numbers

HCF and LCM come straight out of the same prime factorisation

(a) Write 756 as a product of its prime factors.

[2]

(b) Find the smallest positive integer k such that $756k$ is a square number.

$k =$ [1]

Question 2

Recurring decimals and fractions

Show all your working — a bare answer earns nothing on fraction questions

(a) Write $0.\dot{2}1$ as a fraction.

[2]

(b) Work out $2\frac{3}{5} \div 1\frac{1}{4}$. Give your answer as a mixed number in its simplest form.

[2]

Question 3

Negative and fractional indices

Recall squares to 15^2 and cubes of 1, 2, 3, 4, 5 and 10

(a) Find the value of $\left(\frac{16}{81}\right)^{-\frac{3}{4}}$.

[2]

(b) $4^n \times 8 = 2^{11}$. Find the value of n.

$n =$ [2]

Question 4

Standard form

Multiply the numbers, add the powers, then fix the front number so it lies between 1 and 10

Work out $(6 \times 10^{5}) \times (4 \times 10^{-9})$. Give your answer in standard form.

[2]

Question 5

Surds

Rationalise by multiplying top and bottom by the conjugate

(a) Simplify $\sqrt{75} - \sqrt{12}$.

[2]

(b) Rationalise the denominator and simplify $\dfrac{4}{3 - \sqrt{5}}$.

[2]

Question 6

Reverse percentages

The new price is 115% of the original — divide, never subtract 15%

The price of a bicycle increases by 15%. The new price is $92.
Find the original price.

$ ..... [2]

2

Algebra I — manipulation and equations

Video 2 of 6 · Questions 7–11 · 17 marks · E2.2 – E2.5 · 17:50 target runtime

Question 7

Expanding three brackets

Expand two brackets first, then multiply every term by the third

Expand and simplify $(x + 3)(x - 2)(2x + 1)$.

[3]

Question 8

Algebraic fractions

Common denominator first — and put brackets round every numerator

Write $\frac{3}{x-1} - \frac{2}{x+2}$ as a single fraction in its simplest form.

[3]

Question 9

Completing the square

Solutions may be asked for in surd form

(a) Write $x^2 - 6x + 4$ in the form $(x - a)^2 - b$.

[2]

(b) Hence solve $x^2 - 6x + 4 = 0$. Give your answers in surd form.

$x =$ ..... or $x =$ ..... [1]

Question 9 (continued)

Completing the square

(c) Write down the coordinates of the turning point of the graph of $y = x^2 - 6x + 4$.

(..... , ...) [1]

Question 10

Changing the subject

Subject appears twice: clear the fraction, collect the x-terms, factorise

Make x the subject of the formula $y = \dfrac{x + 2}{x - 3}$.

$x =$ ..... [3]

Question 11

Simultaneous equations — one linear, one non-linear

Substitute the line into the curve, then solve the quadratic

Solve the simultaneous equations.

$y = x + 1$

$y = x^2 - 2x - 3$

$x =$ ..... , $y =$ .....

$x =$ ..... , $y =$ ..... [4]

3

Algebra II — sequences, proportion, functions, calculus

Video 3 of 6 · Questions 12–15 · 16 marks · E2.7 – E2.13 · 16:50 target runtime

Question 12

Linear sequences

The common difference is the coefficient of n

Find the *n*th term of this sequence.

5, 12, 19, 26, ...

[2]

Question 13

Direct proportion

Write the equation with k first, find k, then use it

y is directly proportional to the square root of x.

When $x = 9$, $y = 12$.

Find y when $x = 25$.

$y =$ ..... [3]

Question 14

Functions — composite and inverse

gf(x) means do f first, then g

$f(x) = 2x - 3 \qquad g(x) = \dfrac{1}{x+1},\ x \neq -1$

(a) Find f(4).

[1]

(b) Find gf(3).

[1]

Question 14 (continued)

Functions — composite and inverse

$f(x) = 2x - 3$ $\quad g(x) = \dfrac{1}{x+1}, \ x \neq -1$

(c) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... [2]

(d) Solve $ff(x) = 11$.

$x =$ ..... [2]

Question 15

Differentiation and turning points

Turning points: set dy/dx = 0. Nature: use the second derivative

A curve has equation $y = x^3 - 6x^2 + 9x + 2$.

(a) Find $\frac{dy}{dx}$.

[1]

(b) Find the coordinates of the two turning points.

(..... , ...) and (..... , ...) [3]

Question 15 (continued)

Differentiation and turning points

A curve has equation $y = x^3 - 6x^2 + 9x + 2$.

(c) Determine which turning point is a maximum. Show how you decide.

[1]

4

Coordinate geometry and geometry

Video 4 of 6 · Questions 16–19 · 14 marks · E3.4 – E4.7 · 14:50 target runtime

Question 16

Midpoint, length and perpendicular lines

Perpendicular gradients multiply to −1

A is the point $(-2, 5)$ and B is the point $(4, -3)$.

(a) Find the coordinates of the midpoint of AB.

(..... , ...) [1]

(b) Find the length of AB.

..... [2]

Question 16 (continued)

Midpoint, length and perpendicular lines

A is the point $(-2, 5)$ and B is the point $(4, -3)$.

(c) Find the equation of the perpendicular bisector of AB. Give your answer in the form $y = mx + c$.

$y =$ ..... [3]

Question 17

Angles in regular polygons

Exterior angle = 180° − interior angle; the exterior angles add up to 360°

The interior angle of a regular polygon is 156°.
Find the number of sides of this polygon.

[2]

Question 18

Circle theorems

Every angle needs a geometrical reason, in the syllabus wording

A, *B*, *C* and *D* lie on the circle, centre *O*. *TA* is a tangent to the circle at *A*. Angle *ABC* = 68°.

(a) Find angle *AOC*. Give a reason.

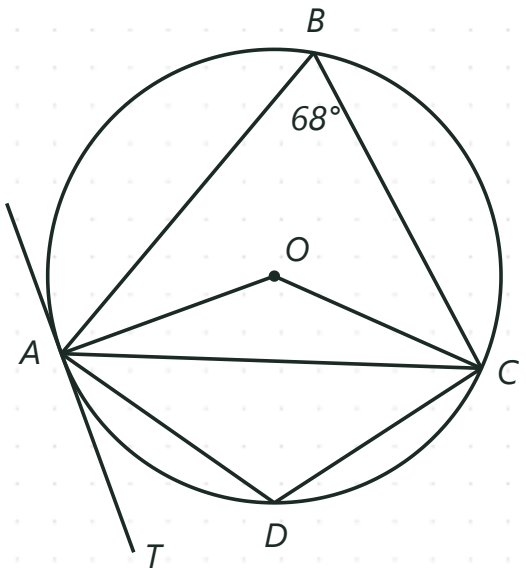
Angle *AOC* = ... because ..... [1]

(b) Find angle *ADC*. Give a reason.

Angle *ADC* = ... because ..... [1]

(c) Find angle *CAT*. Give a reason.

Angle *CAT* = ... because ..... [1]



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Question 19

Similar solids — area and volume

Lengths k : areas k^2 : volumes k^3

Two containers are mathematically similar.
Their surface areas are 45 cm^2 and 80 cm^2.
The volume of the smaller container is 54 cm^3.
Find the volume of the larger container.

..... cm^3 [3]

5

Mensuration, trigonometry, transformations and vectors

Video 5 of 6 · Questions 20–23 · 20 marks · E5.3 – E7.4 · 20:50 target runtime

Question 20

Sectors and cones

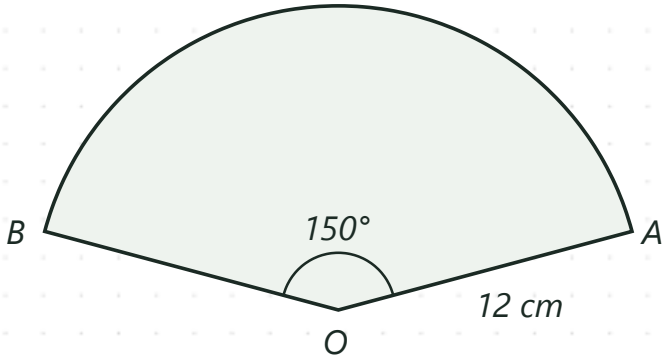
Arc of the sector = circumference of the cone's base. Cone volume is in the formula list

The diagram shows a sector *OAB* of a circle, centre *O*, radius 12 cm and angle 150°.

(a) Find the arc length *AB*. Give your answer in terms of π.

..... cm [2]

(b) Find the area of the sector. Give your answer in terms of π.



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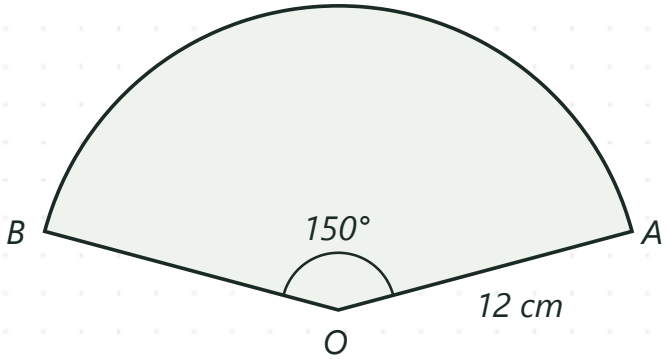
..... cm^2 [2]

Question 20 (continued)

Sectors and cones

The diagram shows a sector *OAB* of a circle, centre *O*, radius 12 cm and angle 150°.

(c) The sector is folded so that *OA* meets *OB*, making a hollow cone.
Find the volume of the cone. Give your answer in terms of π, in its simplest surd form.



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..... cm^3 [3]

Question 21

Exact trigonometric values and trig equations

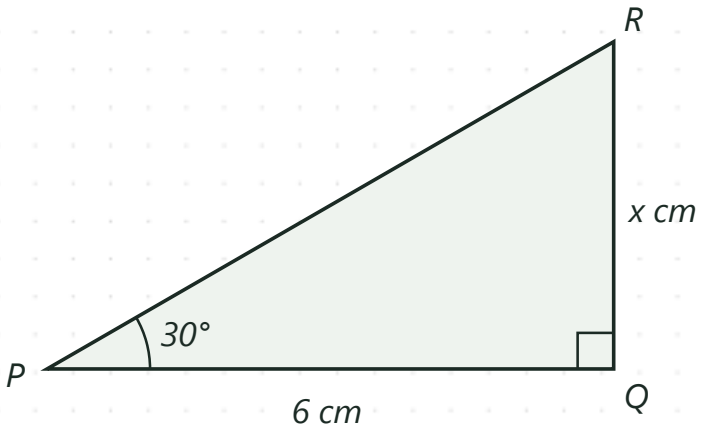
Know sin, cos, tan of 0°, 30°, 45°, 60°, 90° exactly

(a) Write down the exact value of cos 30°.

.....

[1]

(b) In triangle PQR, angle $PQR = 90°$, angle $QPR = 30°$ and $PQ = 6$ cm.
Find the exact value of x.



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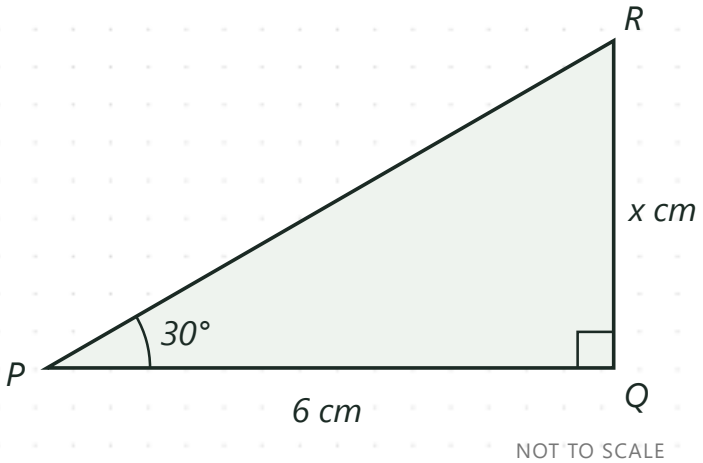
$x =$ .....

[2]

Question 21 (continued)

Exact trigonometric values and trig equations

(c) Solve $2 \sin x + \sqrt{3} = 0$ for $0° \leq x \leq 360°$.



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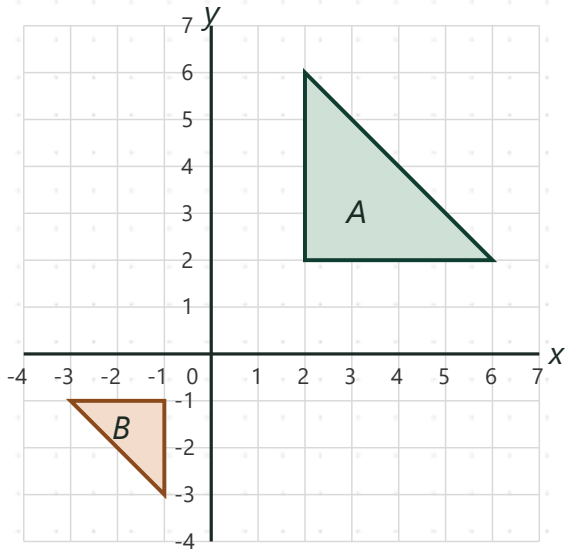
$x =$ ..... or $x =$ ..... [3]

Question 22

Transformations — negative scale factor

Three facts for an enlargement: the word, the scale factor, the centre

Describe fully the **single** transformation that maps triangle *A* onto triangle *B*.



[3]

Question 23

Vector geometry

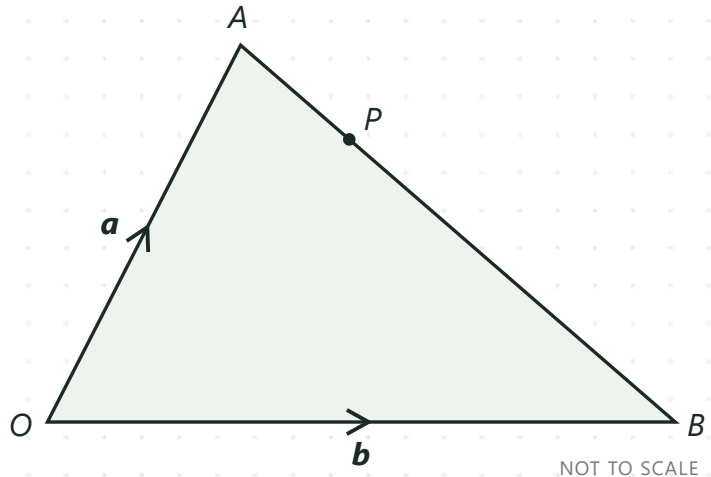
Route through known points; collinear means one vector is a multiple of the other plus a shared point

$\overrightarrow{OA}$ = **a** and $\overrightarrow{OB}$ = **b**. *P* lies on *AB* such that *AP* : *PB* = 1 : 3.

(a) Find $\overrightarrow{AB}$ in terms of **a** and **b**.

[1]

(b) Find $\overrightarrow{OP}$ in terms of **a** and **b**, in its simplest form.



[2]

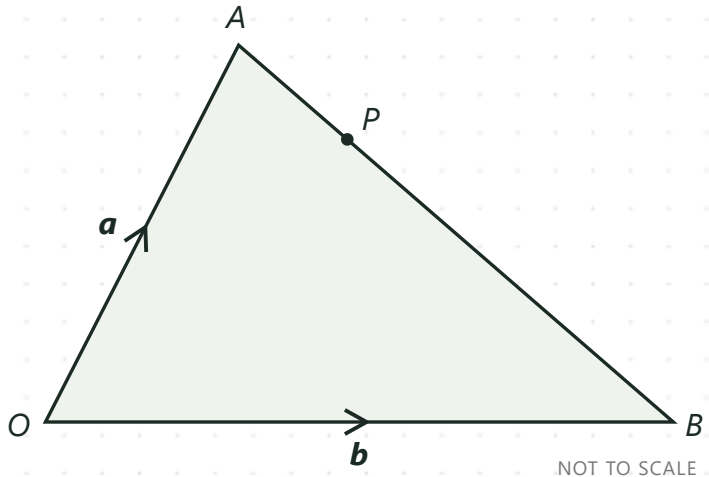
Question 23 (continued)

Vector geometry

$\overrightarrow{OA}$ = **a** and $\overrightarrow{OB}$ = **b**. *P* lies on *AB* such that *AP* : *PB* = 1 : 3.

(c) $\overrightarrow{OQ}$ = 3**a** + **b**.

Explain why *O*, *P* and *Q* lie on a straight line.



NOT TO SCALE

[1]

6

Probability and statistics

Video 6 of 6 · Questions 24–26 · 14 marks · E8.3 – E9.7 · 14:50 target runtime

Question 24

Combined events — without replacement

Second pick: one fewer in the bag, top and bottom

A bag contains 4 red counters and 6 green counters. Two counters are taken at random, without replacement.

(a) Find the probability that both counters are green.

[2]

(b) Find the probability that the two counters are different colours.

[2]

Question 25

Grouped data — estimated mean and histogram

Estimated mean uses midpoints. Frequency density = frequency ÷ class width

The table shows the times, t minutes, that 40 people waited at a clinic.

(a) Calculate an estimate of the mean waiting time.

Time (t minutes)	Frequency
$0 < t \leq 10$	8
$10 < t \leq 20$	12
$20 < t \leq 40$	16
$40 < t \leq 60$	4

..... min [3]

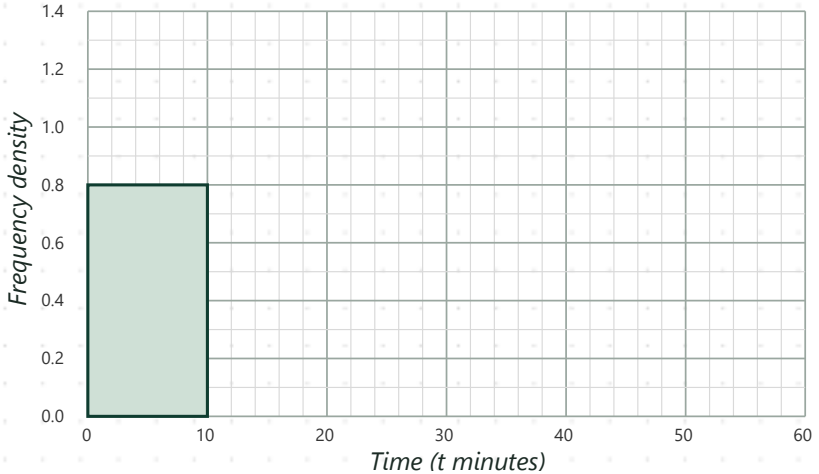
Question 25 (continued)

Grouped data — estimated mean and histogram

The table shows the times, t minutes, that 40 people waited at a clinic.

(b) Complete the histogram to show the information in the table.

Time (t minutes)	Frequency
$0 < t \leq 10$	8
$10 < t \leq 20$	12
$20 < t \leq 40$	16
$40 < t \leq 60$	4



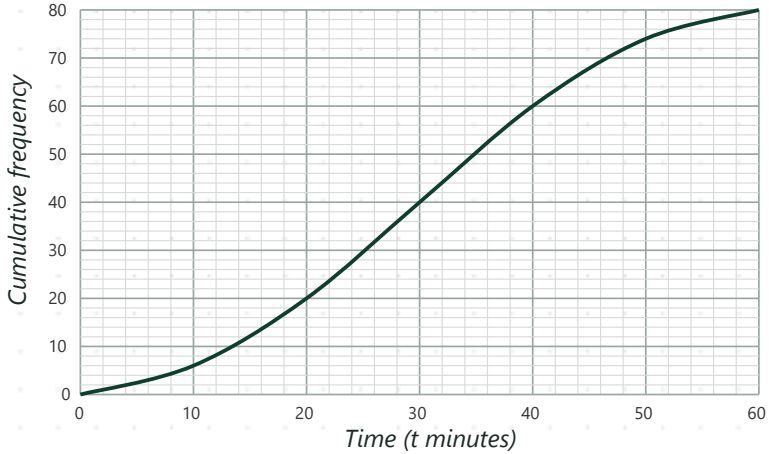
[3]

Question 26

Cumulative frequency

Median at n ÷ 2, quartiles at n ÷ 4 and 3n ÷ 4 — read across, then down

The cumulative frequency diagram shows the times, *t* minutes, that 80 students spent on homework one evening.



(a) Use the diagram to find an estimate of the median.

..... min [1]

(b) Use the diagram to find an estimate of the interquartile range.

..... min [2]

(c) Use the diagram to find an estimate of the number of students who spent more than 50 minutes.

..... [1]

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Answer key 1 of 4

Every answer has been worked through by hand. Grey line = how the marks are earned, in Cambridge style: M method · A accuracy · B independent · FT follow-through.

Q	CODE	TOPIC	MK	ANSWER
01	E1.1	Prime factors and square numbers	3	$756 = 2 \times 2 \times 3 \times 3 \times 3 \times 7 = \mathbf{2^2 \times 3^3 \times 7}$ | $k = 3 \times 7 = \mathbf{21}$ ($756 \times 21 = 15\,876 = 126^2$) (a) B2 – B1 for 2, 2, 3, 3, 3, 7 listed or a correct factor tree · (b) B1
02	E1.4 / E1.6	Recurring decimals and fractions	4	$x = 0.2111\ldots$, $100x - 10x = 21.1\ldots - 2.1\ldots = 19 \rightarrow \mathbf{19/90}$ | $13/5 \times 4/5 = 52/25 = \mathbf{2\ 2/25}$ (a) M1 for 100x – 10x or 90x = 19 · A1 · (b) M1 for 13/5 × 4/5 · A1
03	E1.7 / E2.4	Negative and fractional indices	4	$(81/16)^{3/4} = (3/2)^3 = \mathbf{27/8}$ | $2^{2n} \times 2^3 = 2^{11} \rightarrow 2n + 3 = 11 \rightarrow \mathbf{n = 4}$ (a) M1 for 81/16 flipped or 3/2 seen · A1 · (b) M1 for $2^{2n} \times 2^3$ or 2n + 3 = 11 · A1
04	E1.8	Standard form	2	$24 \times 10^{-4} = \mathbf{2.4 \times 10^{-3}}$ M1 for 24×10^{-4} or 2.4×10^{k} · A1
05	E1.18	Surds	4	$5\sqrt{3} - 2\sqrt{3} = \mathbf{3\sqrt{3}}$ | $4(3 + \sqrt{5}) / (9 - 5) = \mathbf{3 + \sqrt{5}}$ (a) M1 for 5√3 or 2√3 · A1 · (b) M1 for × (3 + √5)/(3 + √5) · A1
06	E1.13	Reverse percentages	2	$92 \div 1.15 = 9200 \div 115 = \mathbf{\$80}$ (not 92×0.85) M1 for 92 ÷ 1.15 or 1.15x = 92 · A1
07	E2.2	Expanding three brackets	3	$(x^2 + x - 6)(2x + 1) = \mathbf{2x^3 + 3x^2 - 11x - 6}$ M1 for x² + x – 6 · M1 for 3 of 4 terms correct · A1

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Answer key 2 of 4

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Q	CODE	TOPIC	MK	ANSWER
08	E2.3	Algebraic fractions	3	$[3(x+2)-2(x-1)] / [(x-1)(x+2)] = \mathbf{(x+8) / ((x-1)(x+2))}$ M1 for common denominator (x − 1)(x + 2) · M1 for 3(x + 2) − 2(x − 1) · A1
09	E2.5 / E2.11	Completing the square	4	(a) $\mathbf{(x-3)^2-5}$ | (b) $\mathbf{x = 3 \pm \sqrt{5}}$ | (c) $\mathbf{(3, -5)}$ (a) B1 a = 3 · B1 b = 5 · (b) B1 FT · (c) B1 FT
10	E2.5	Changing the subject	3	$xy - 3y = x + 2 \rightarrow xy - x = 3y + 2 \rightarrow \mathbf{x = (3y+2) / (y-1)}$ M1 for clearing the fraction · M1 for collecting x-terms and factorising · A1
11	E2.5	Simultaneous equations — one linear, one non-linear	4	$x^2 - 3x - 4 = 0 \rightarrow (x-4)(x+1) = 0 \rightarrow \mathbf{x = 4, y = 5}$ and $\mathbf{x = -1, y = 0}$ M1 for equating · M1 for x² − 3x − 4 = 0 solved · A1 both x · A1 both y
12	E2.7	Linear sequences	2	difference $7 \rightarrow 7n + 5 - 7 \rightarrow \mathbf{7n - 2}$ B2 — B1 for 7n + k
13	E2.8	Direct proportion	3	$y = k\sqrt{x} \rightarrow 12 = 3k \rightarrow k = 4 \rightarrow y = 4 \times 5 = \mathbf{20}$ M1 for y = k√x · A1 for k = 4 · A1
14	E2.13	Functions — composite and inverse	6	(a) **5** | (b) f(3) = 3, g(3) = **1/4** | (c) $\mathbf{(x+3)/2}$ | (d) $2(2x-3) - 3 = 4x - 9 = 11 \rightarrow \mathbf{x = 5}$ (a) B1 · (b) B1 · (c) M1 for x = 2y − 3 or y + 3 = 2x · A1 · (d) M1 for 2(2x − 3) − 3 = 11 · A1

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Answer key 3 of 4

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Q	CODE	TOPIC	MK	ANSWER
15	E2.12	Differentiation and turning points	5	(a) **$3x^2 - 12x + 9$** | (b) $3(x - 1)(x - 3) = 0 \rightarrow$ **(1, 6)** and **(3, 2)** | (c) $d^2y/dx^2 = 6x - 12 = -6$ at $x = 1 \rightarrow$ **(1, 6) is the maximum** (a) B1 · (b) M1 for dy/dx = 0 · A1 both x · A1 both points · (c) B1 with a valid reason
16	E3.4 – E3.7	Midpoint, length and perpendicular lines	6	(a) **(1, 1)** | (b) $\sqrt{(6^2 + 8^2)} =$ **10** | (c) $m_{AB} = -8/6 = -4/3 \rightarrow$ perpendicular 3/4 through (1, 1) $\rightarrow$ ***y = ¾x + ¼*** (a) B1 · (b) M1 for $6^2 + 8^2$ · A1 · (c) M1 gradient AB · M1 perpendicular gradient · A1
17	E4.6	Angles in regular polygons	2	exterior = 180 − 156 = 24° → 360 ÷ 24 = **15** M1 for 180 − 156 or 360 ÷ 24 · A1
18	E4.7	Circle theorems	3	(a) **136°** angle at centre is twice the angle at the circumference | (b) **112°** opposite angles of a cyclic quadrilateral add to 180° | (c) **68°** alternate segment theorem B1 each — the angle AND the reason are both needed for the mark
19	E4.4	Similar solids — area and volume	3	areas 9 : 16 → lengths 3 : 4 → volumes 27 : 64 → 54 × 64/27 = **128 cm³** M1 for √(80/45) = 4/3 · M1 for $(4/3)^3 \times 54$ · A1
20	E5.3 / E5.4	Sectors and cones	7	(a) 150/360 × 24π = **10π** | (b) 150/360 × 144π = **60π** | (c) $2\pi r = 10\pi \rightarrow r = 5$, $h = \sqrt{(144 - 25)} = \sqrt{119} \rightarrow \frac{1}{3}\pi(25)\sqrt{119} =$ **$(25\sqrt{119}/3)\pi$ cm³** (a) M1 · A1 · (b) M1 · A1 · (c) M1 for r = 5 · M1 for h = √119 · A1
21	E6.2 – E6.4	Exact trigonometric values and trig equations	6	(a) **$\sqrt{3}/2$** | (b) $x = 6 \tan 30° = 6/\sqrt{3} =$ **$2\sqrt{3}$** | (c) $\sin x = -\sqrt{3}/2 \rightarrow$ **240° and 300°** (a) B1 · (b) M1 for 6 tan 30° · A1 · (c) B1 for sin x = −√3/2 · B1 240° · B1 300°

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Answer key 4 of 4

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Q	CODE	TOPIC	MK	ANSWER
22	E7.1	Transformations — negative scale factor	3	**Enlargement**, **scale factor −½**, **centre (0, 0)** — one transformation only B1 enlargement · B1 scale factor −½ · B1 centre (0, 0) — two transformations scores 0
23	E7.2 – E7.4	Vector geometry	4	(a) **b − a** | (b) a + ¼(b − a) = **¾a + ¼b** | (c) OQ = 3a + b = 4(¾a + ¼b) = **4 *OP***: parallel and share point O (a) B1 · (b) M1 for a correct route, e.g. a + ¼(b − a) · A1 · (c) B1
24	E8.3	Combined events — without replacement	4	(a) 6/10 × 5/9 = **1/3** | (b) 2 × 4/10 × 6/9 = **8/15** (a) M1 for 6/10 × 5/9 · A1 · (b) M1 for 4/10 × 6/9 + 6/10 × 4/9 · A1
25	E9.3 / E9.7	Grouped data — estimated mean and histogram	6	(a) Σfx = 8×5 + 12×15 + 16×30 + 4×50 = 900 → 900 ÷ 40 = **22.5 min** | (b) bars at FD **1.2** (10–20), **0.8** (20–40), **0.2** (40–60) (a) M1 for midpoints · M1 for Σfx ÷ 40 · A1 · (b) B1 for each correct bar
26	E9.6	Cumulative frequency	4	(a) **30 min** (CF 40) | (b) UQ 40 − LQ 20 = **20 min** | (c) 80 − 74 = **6** (a) B1 · (b) B1 for UQ 40 or LQ 20 · B1 · (c) B1